

Assignment 5: Data Visualization

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the NEON_NIWO_Litter_mass_trap_Processed.csv version, again from the Processed_KEY folder).
2. Make sure R is reading dates as date format; if not change the format to date.

```
#1
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    4.0.0      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(here)
```

```
## here() starts at /Users/carinahuang/EDE2025
```

```
library(cowplot)
```

```
##  
## Attaching package: 'cowplot'  
##  
## The following object is masked from 'package:lubridate':  
##  
## stamp
```

```
getwd()
```

```
## [1] "/Users/carinahuang/EDE2025"
```

```
peterpaul1 <- read.csv(here("Data/Raw/Processed_KEY/NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv"))
```

```
niwo <- read.csv(here("Data/Raw/Processed_KEY/NEON_NIWO_Litter_mass_trap_Processed.csv"),  
  stringsAsFactors = TRUE)
```

```
#2
```

```
peterpaul1$sampldate <- ymd(peterpaul1$sampldate)  
niwo$collectDate <- ymd(niwo$collectDate)
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```
#3
```

```
mytheme <- theme_classic(base_size = 14) +  
  theme(axis.text = element_text(color = "black"),  
    axis.title = element_text(family = 'Avenir'),  
    plot.title = element_text(family = 'Avenir', size = 18),  
    legend.position = "top",  
    legend.title = element_text(family = 'Avenir')  
  )
```

Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (tp_ug) by phosphate (po4), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

```
#4
plot4 <- ggplot(peterpaul1, aes(x=tp_ug, y=po4, color = lakenam)) +
  geom_point() +
  geom_smooth(method = lm, se=FALSE) +
  ylim(0,60) +
  mytheme +
  labs(title = "Peter and Paul Lake phosphorus and phosphate level") +
  labs(x= "time")
```

5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis title in your theme to `element_blank()` removes the axis title (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different sizes when combined using `cowplot`.

```
#5

plot_temp <- ggplot (peterpaul1, aes(x=month, y=temperature_C, color = lakenam)) +
  geom_boxplot() +
  aes(x=factor(
    month,
    levels = 1:12,
    labels = month.abb
  )) +
  labs(x= "Month") +
  theme(legend.position = "none")
plot_tp <- ggplot (peterpaul1, aes(x=month, y=tp_ug, color = lakenam)) +
  geom_boxplot() +
  aes(x=factor(
    month,
    levels = 1:12,
    labels = month.abb
  )) +
  labs(x= "Month") +
  theme(legend.position = "none")

plot_tn <- ggplot (peterpaul1, aes(x=month, y=tn_ug, color = lakenam)) +
  geom_boxplot() +
  aes(x=factor(
    month,
    levels = 1:12,
```

```

    labels = month.abb
  ))+
  labs(x= "Month")

legend <- get_legend(plot_tn)

```

```

## Warning: Removed 21583 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```

```

plot_grid(plot_temp, plot_tp, plot_tn+theme(legend.position = "none"),legend,
  nrow = 4, align = 'h')

```

```

## Warning: Removed 3566 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```

```

## Warning: Removed 20729 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```

```

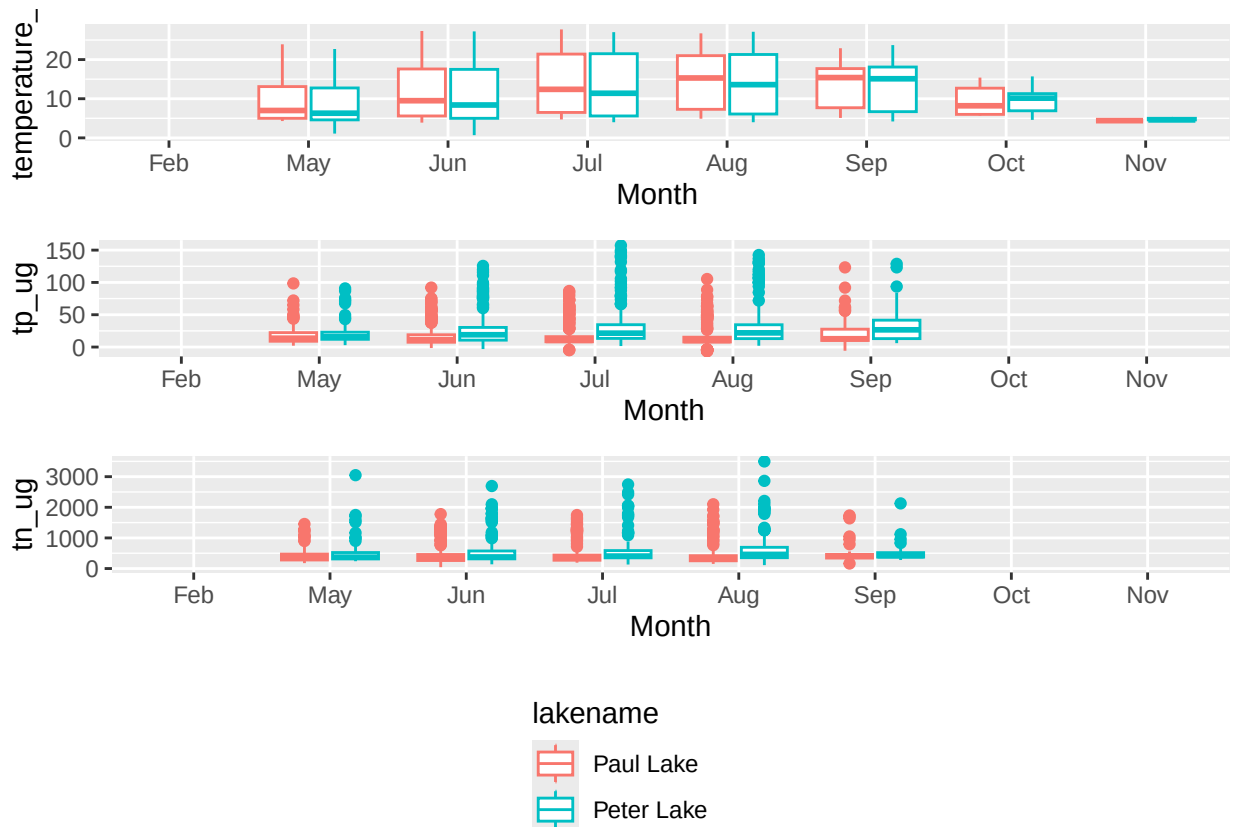
## Warning: Removed 21583 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```

```

## Warning: Graphs cannot be horizontally aligned unless the axis parameter is
## set. Placing graphs unaligned.

```



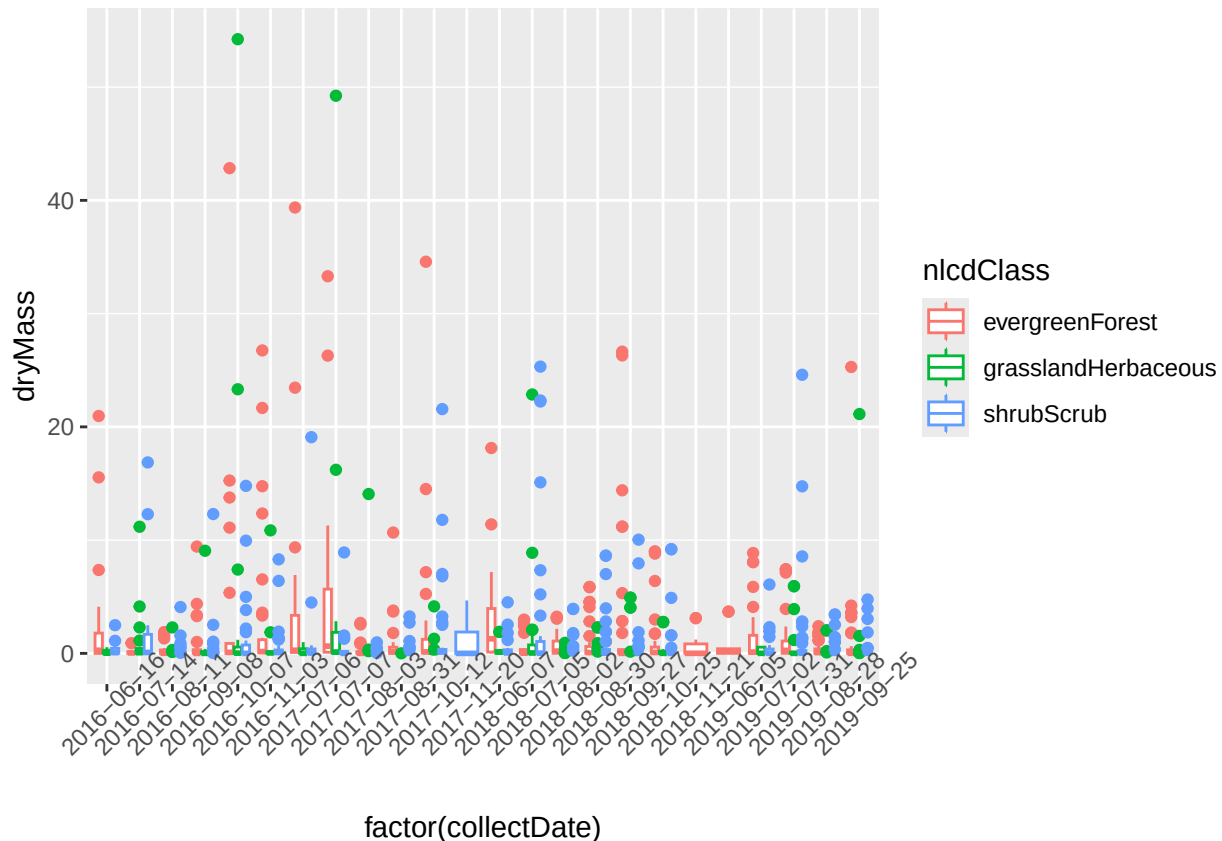
Question: What do you observe about the variables of interest over seasons and between lakes?

Answer: during warmer months (seasons), the lake temperature is higher, and the different nutrient levels are also higher compared to colder months.

6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the “Needles” functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

```
#6
plot_needle <- filter(niwo, functionalGroup == "Needles")
drymass_plot <- ggplot(niwo, aes(x=factor(collectDate),
                                   y=dryMass, color = nlcdClass))+
  geom_boxplot()+
  theme(axis.text.x = element_text(angle =45))

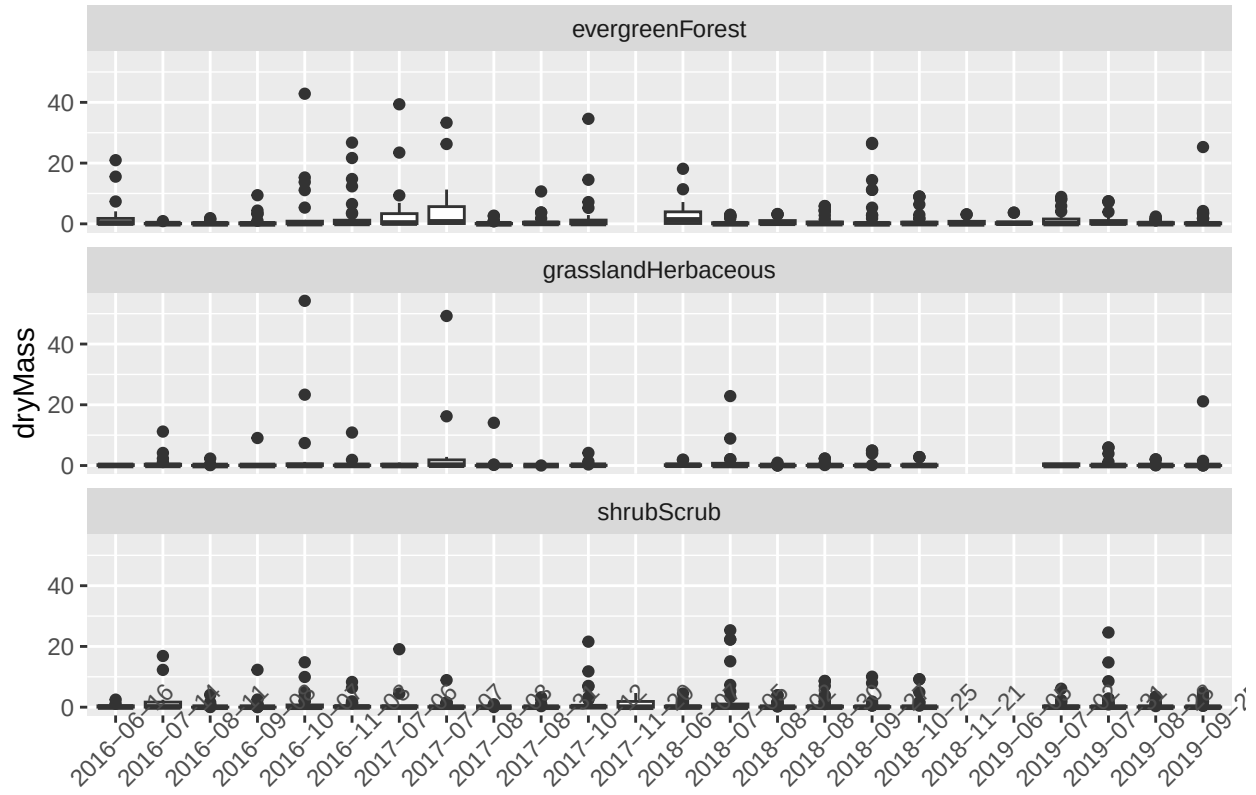
print(drymass_plot)
```



```
#7
drymass_plot_facet <- ggplot(niwo, aes(x=factor(collectDate), y=dryMass))+
  geom_boxplot()+
  facet_wrap(facets = vars(nlcdClass),
             nrow =3)+
  theme(axis.text.x = element_text(angle =45, vjust = 1))+
```

```
labs(x=NULL)
```

```
print(drymass_plot_facet)
```



Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

Answer: I think plot 6 is more effective because it is easier to compare the dry mass of the different categories on the same plot collected on the same day.